

MANTHAN JAIN

Mumbai, Maharashtra, India

✉ manthanmaheshjain@gmail.com | 📞 +91 -8291 682893

LinkedIn: www.linkedin.com/in/manthan-jain-855413369 | GitHub: github.com/ManthanJain-bit

PROFESSIONAL SUMMARY

"High-achieving Computer Engineering student with a **9.68/10 GPA** and a specialized focus on **Artificial Intelligence and Real-Time Systems Optimization**. Demonstrated expertise in developing low-latency architectures in C++ and architecting end-to-end AI pipelines for facial analysis and recommendation systems. Proficient in leveraging mathematical foundations like **Linear Algebra and Statistics** to improve computational efficiency and algorithmic modeling. Certified in **Data Structures and Digital Image Processing**, with a proven ability to bridge complex theory with scalable, high-performance software solutions."

EDUCATION

Bachelor of Engineering in Computer Engineering

2024 – 2028

Vidyalankar Institute of Technology, Mumbai

GPA: 9.68/10

Relevant CourseWork

- Data Structures & Algorithms
- Operating Systems
- Engineering Mathematics
- Linear Algebra
- Probability & Statistics

RESEARCH INTERESTS

Artificial Intelligence (Efficient ML Systems), Real-Time Systems Optimization, Computational Modeling, Algorithmic Efficiency, Scalable Distributed Systems

RESEARCH & TECHNICAL EXPERIENCE

Game Systems Development Intern

Cascade Tech / May 2025 – June 2025

- Designed and implemented low-latency event-driven systems in C++, focusing on computational efficiency and scalable architecture design.
- Developed and optimized physics-based collision detection algorithms, analyzing time complexity and improving real-time response accuracy.
- Enhanced runtime performance through memory profiling, bottleneck analysis, and algorithmic optimization within Unreal Engine.
- Engineered modular system components following version-controlled, collaborative development workflows using Git.
- Applied algorithmic modeling techniques to scoring and interaction logic, improving system responsiveness and deterministic behavior.

PROJECTS

Real-Time Interactive Simulation System (C++ & Unreal Engine)

- Architected deterministic simulation framework emphasizing low-latency execution.
- Designed optimized spatial collision detection mechanisms reducing computational overhead.
- Conducted iterative performance profiling and refactoring to enhance runtime efficiency.

Embedded Gesture-Controlled Robotic System (Arduino + Python)

- Developed embedded control architecture for obstacle-aware robotic navigation.
- Implemented real-time gesture classification pipeline integrating microcontroller-level programming.
- Applied signal processing logic for structured decision modeling.

Database-Driven Application Systems (Java & C++ with MySQL)

- Designed normalized relational schemas to optimize query performance and data consistency.
- Implemented modular backend systems emphasizing scalability and structured architecture design.

Vision India – AI-Driven Face-Based Recommendation System

- Developed an end-to-end facial analysis pipeline leveraging landmark detection and geometric ratio modeling for structured face-shape classification.
- Implemented rule-based and feature-mapped recommendation logic to optimize frame matching accuracy under constrained mobile environments.
- Conducted performance profiling to balance inference latency, model complexity, and memory utilization.
- Designed distributed database architecture supporting real-time transactional consistency and offline synchronization.
- Integrated analytics-driven feedback loop to iteratively refine recommendation heuristics.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, System Design

Technical Areas: Real-Time Systems, Algorithm Optimization, Collision Detection, Performance Tuning

Tools & Technologies: Git, MySQL, Unreal Engine, Firebase, SQLite

CERTIFICATIONS

- • Data Structures & Algorithms – NPTEL ▪ • Digital Image Processing – NPTEL ▪ • Java Programming – NPTEL
- • Introduction to Python – Coursera ▪ • Diploma in Robotics – Alison